

Tanvir Ahmed Khan

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[Google Scholar](#) • [GitHub](#) • [LinkedIn](#)

Education

North South University

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

2021 – 2025

- **CGPA: 3.80/4.00** | *Summa Cum Laude*
- Relevant Coursework: Deep Learning, Pattern Recognition, Natural Language Processing, Numerical Analysis

Research Interests

Multimodal Learning and Optimization; 3D Computer Vision, Neural Rendering, and Gaussian Splatting; Speech-Driven Generative Models and Talking-Head Synthesis; Predictive Representation Learning, World Models, and Long-Video Understanding; Memory-Augmented AI; Low-Resource NLP and Context-Aware Language Understanding.

Publications

Peer-Reviewed Publications and Accepted Papers

- [1] Mahir Shahriar Tamim, Md. Samiul Alim, **Tanvir Ahmed Khan**, Maisha Rahman, and Md Musfique Anwar. “Ensemble and Temporal Feature-Based Framework for Rainfall Classification in Bangladesh.” *PLOS ONE*, 21(3): e0342646, 2026. [DOI](#).
- [2] Mahir Shahriar Tamim, Sharjil Khan, Md. Samiul Alim, **Tanvir Ahmed Khan**, Shafin Rahman, and Nabeel Mohammed. “CAT-GS: Balanced Multimodal Learning via Calibrated Gating and Fusion Surgery.” *Neurocomputing*, accepted for publication, 2026. [arXiv](#).
- [3] Md. Samiul Alim, Mahir Shahriar Tamim, **Tanvir Ahmed Khan**, Sharjil Khan, Rafia Ferdous Duti, Shahriyar Zaman Ridoy, and Mohammad Ali Moni. “Can LLMs Follow the Pulse of a Crisis? Evaluating Crisis Sentiment in Bangladesh’s July Uprising.” *AACL-IJCNLP 2026*, Main Conference, accepted, 2026.
- [4] Md. Samiul Alim, Mahir Shahriar Tamim, Maisha Rahman, **Tanvir Ahmed Khan**, and Md Musfique Anwar. “When a Nation Speaks: Machine Learning and NLP in People’s Sentiment Analysis During Bangladesh’s 2024 Mass Uprising.” In *2025 28th International Conference on Computer and Information Technology (ICCIT)*, pp. 413–418, IEEE, 2025. [DOI](#).

Research Experience

Machine Intelligence Lab (MILab) | North South University

Mar 2025 – Present

- **Research Assistant.** Advisors: [Dr. Nabeel Mohammed](#) and [Dr. Shafin Rahman](#).
- Conduct research in multimodal learning, low-resource natural language processing, and generative talking-head systems, contributing to peer-reviewed journal and conference work.
- Developed distributed multi-GPU inference pipelines for talking-avatar generation, including chunked execution for computationally intensive generative workloads.
- Integrated NVIDIA Audio2Face with Three.js and investigated ARKit blend-shape recovery through linear optimization for real-time avatar animation.
- Investigated facial geometry and spatio-temporal motion representations for generative talking-head systems, including FLAME/3DMM-based tracking and 3D Gaussian Splatting.
- Developing deployment-oriented speech and avatar pipelines targeting NVIDIA Jetson AGX Orin.

Selected Research Projects

Speech2Avatar: Speech-Driven 3D Talking-Head Generation	Mar 2026 – Present
Research on speech-conditioned facial geometry and Gaussian-Splatting-based talking-head synthesis, including person-specific facial modeling and deployment-oriented speech/avatar components.	
Predictive Representation Learning for Long-Video Understanding	2026 – Present
Ongoing research on JEPA-style predictive video representations and hierarchical memory for efficient long-horizon video understanding, retrieval, and reasoning.	
Saliency Detection via Knowledge Distillation	Aug 2024 – Dec 2024
Knowledge-distillation prototype using a Peekaboo/DINO teacher and UKAN student with mask-supervised and teacher–student objectives.	GitHub
DynApex: Dynamic CPU/GPU Parallelization	Aug 2024 – Dec 2024
C++ prototype for dependency and cycle analysis, pragma injection, and OpenMP-based CPU and target execution.	GitHub

Industry Experience

Tuctuc Spain (Remote)	Jan 2022 – Jun 2026
• Junior Software Engineer. Maintained and extended Odoo-based ERP systems and developed cross-platform mobile applications using React Native.	

Technical Skills

Machine Learning & Deep Learning: PyTorch, Hugging Face Transformers, Diffusion Models, Variational Autoencoders, Knowledge Distillation, PEFT/LoRA	
3D Vision & Neural Rendering: 3D Gaussian Splatting, FLAME, 3D Morphable Models (3DMM), Talking-Head Generation, Facial Motion Modeling	
Multimodal & Video Learning: Vision-Language Models, CLIP, Multimodal Fusion, Video Representation Learning, Predictive Representation Learning	
NLP, Retrieval & Memory: Large Language Models, Retrieval-Augmented Generation, FAISS, Knowledge Graphs, SQL, Sentiment Analysis, Topic Modeling	
Speech & Avatar Systems: Speech-to-Speech Models, Text-to-Speech, Mel-Spectrogram Analysis, NVIDIA Audio2Face, ARKit Blend Shapes, Lip-Synchronization Systems	
Systems & Deployment: Multi-GPU Inference, CUDA, Docker, Linux, RunPod, NVIDIA Jetson AGX Orin, FastAPI, Git	
Programming: Python, C++, JavaScript, React Native	
Languages: Bangla (Native), English (C1)	

Awards & Honors

Academic Scholarship , North South University	2023
Preliminary Qualifier , ICPC Asia Dhaka Regional	2021